FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2019

PAPER – 1

TEST DATE: 28-04-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C

Sol. $a_t = 2\ m/s^2$

$$S = ut + \frac{1}{2}a_t t^2 = 0 + \frac{1}{2}\times 2t^2 = t^2$$

$\therefore S = t^2$

$$v^2 = u^2 + 2a_t S = 2\times 2S = 4S$$

$\therefore a_n = t^4 = S^2$

$a_n = S^2$

$$\therefore\ a = \sqrt{a_t^2 + a_n^2} = \sqrt{4 + S^4}$$

At S = 2m

$$a = \sqrt{4+16} = 2\sqrt{5}\ m/s^2$$

Radius of curvature of its trajectory

$$r = \frac{v^2}{a_n} = \frac{4S}{S^2} = \frac{4}{S}$$

At S = 1m, r = 4m,

2. A, D

Sol. $\omega_0 R\cot\theta = v\sin 2\theta$

$$\omega_0 R\frac{\cos\theta}{\sin\theta} = 2v\sin\theta\cos\theta$$

$$\omega_0 R = 2v\sin^2\theta$$

Just after release

$$\alpha_0 R = 2a\sin^2\theta$$

$$\alpha_0 R = 2a\times\frac{9}{25} \text{ (since, } \theta = 37°\text{, initially)}$$

$$\alpha_0 R = \frac{18a}{25} \quad \ldots(i)$$

$N_1 \sin 2\theta - f_s = ma$ …(ii)

$$f_s R = \frac{2}{5}mR^2\alpha$$

$$f_s = \frac{2}{5}ma \quad \ldots(iii)$$

From (ii) and (iii), we get

$$N_1 \sin 2\theta = \frac{7}{5}ma$$

$$N_1\frac{24}{25} = \frac{7}{5}ma \Rightarrow N_1 = 3.5\ a \quad \ldots(iv)$$

For the rod,

$$m_0 g\frac{\ell}{2}\cos 2\theta - N_1 R\cot\theta = \frac{m_0\ell^2}{3}\alpha_0$$

$$\frac{m_0 g}{2}\times\frac{7}{25} - N_1\times 0.5\times\frac{4}{3} = \frac{m_0}{3}\alpha_0 = \frac{m_0}{3}\times\frac{36a}{25}$$

$$3.5 - N_1\frac{2}{3} = 1.2\ a$$

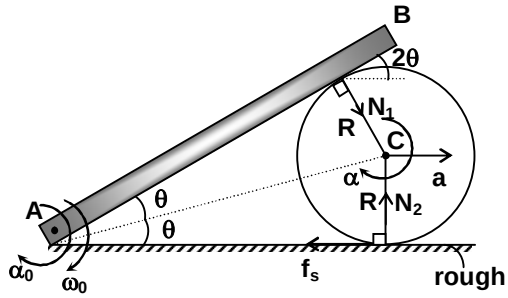
$$3.5 - 3.5a\times\frac{2}{3} = 1.2a$$

$$\Rightarrow 1.2a + \frac{7a}{3} = 3.5$$

$$a = \frac{10.5}{10.6} \approx 1\ m/s^2$$

From (iv)

$N_1 = 3.5\ a = 3.5 \times 1 = 3.5$ N



a = αR

3. A, C

Sol. $$\frac{d^2x}{dt^2} = -\frac{GMr\cos\theta}{R^3} = -\left(\frac{GM}{R^3}\right)x = -\left(\frac{g_0}{R}\right)x$$

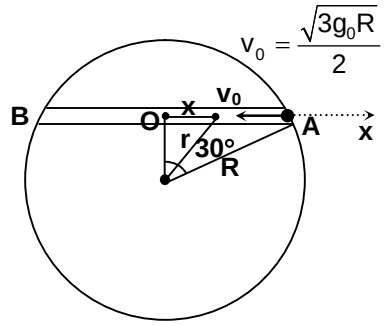
$$\omega = \sqrt{\frac{g_0}{R}}$$

Now, $x = A\sin(\omega t + \alpha)$

at $t = 0$, $x = +R/2$

$$\frac{R}{2} = A\sin\alpha \quad \ldots(i)$$

$V_x = \omega A\cos(\omega t + \alpha)$



at t = 0, $v_x = -\frac{\sqrt{3g_0R}}{2}$

$-\frac{\sqrt{3g_0R}}{2} = \sqrt{\frac{g_0}{R}}A\cos\alpha$

$-\frac{\sqrt{3}}{2}R = A\cos\alpha$ …(ii)

Solving (i) and (ii), we get

$A = R,\ \alpha = \frac{5\pi}{6}$

$\therefore\ x = R\sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$

$-\frac{R}{2} = R\sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$

$\frac{7\pi}{6} = \sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}$

$\therefore\ t = \frac{\pi}{3}\sqrt{\frac{R}{g_0}}$

$v_{max} = \omega A$

$v_{max} = \sqrt{\frac{g_0}{R}}R = \sqrt{g_0R}$

4. A, B, D

Sol. $A_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)A_i = \left(\frac{\sqrt{\frac{F}{\mu_2}} - \sqrt{\frac{F}{\mu_1}}}{\sqrt{\frac{F}{\mu_2}} + \sqrt{\frac{F}{\mu_1}}}\right)A_i = \left(\frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}}\right)A_i = \left(\frac{3-1}{3+1}\right)3 = 1.5\,\text{cm}$

$A_t = \left(\frac{2v_2}{v_2 + v_1}\right)A_i = \left(\frac{2\sqrt{\frac{F}{\mu_2}}}{\sqrt{\frac{F}{\mu_2}} + \sqrt{\frac{F}{\mu_1}}}\right)A_i = \left(\frac{2A_i}{1 + \sqrt{\frac{\mu_2}{\mu_1}}}\right)_i = \left(\frac{2\times 3}{1 + \frac{1}{3}}\right) = 4.5\,\text{cm}$

Now, $\frac{P_r}{P_i} = \left(\frac{A_r}{A_i}\right)^2 = \left(\frac{1.5}{3}\right)^2 = \frac{1}{4}$

$\therefore$ The percentage of incident power reflected to the first string = $\frac{1}{4}\times 100 = 25\,\%$

The percentage of incident power transmitted to the second string = 100 – 25 = 75%

5. B, C

Sol. The electrostatic force between the hemispherical shells is

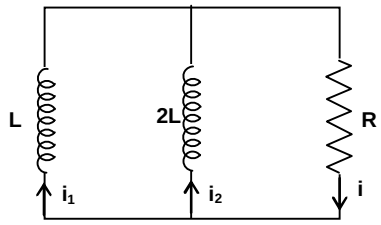
$F = \frac{\sigma_1\sigma_2\pi R^2}{2\varepsilon_0} = \frac{Q}{2\pi R^2}\frac{2Q}{8\pi R^2}\frac{\pi R^2}{2\varepsilon_0} = \frac{Q^2}{16\pi\varepsilon_0 R^2}$

The net electric potential at the centre 'O' of the sells is

$V_0 = \frac{KQ}{R} + \frac{2KQ}{2R} = \frac{2KQ}{R} = \frac{Q}{2\pi\varepsilon_0 R}$

6. A, D

Sol. $L\frac{di_1}{dt} = 2L\frac{di_2}{dt}$



$L\int_{I_0}^{i_1} di_1 = 2L\int_{I_0}^{i_2} di_2$

$i_1 - I_0 = 2(i_2 - I_0)$

$2i_2 - i_1 = I_0$ …(i)

Now, $L\frac{di_1}{dt} + iR = 0$

$L\frac{di_1}{dt} + (i_1 + i_2)R = 0$

$L\frac{di_1}{dt} + \left(\frac{3i_1 + I_0}{2}\right)R = 0$

$\int_{I_0}^{i_1} \frac{di_1}{(3i_1 + I_0)} = -\frac{R}{2L}\int_0^t dt \Rightarrow \ell n\left(\frac{3i_1 + I_0}{4I_0}\right) = -\frac{3Rt}{2L}$

$\therefore\ i_1 = \frac{I_0}{3}\left(4e^{-\frac{3Rt}{2L}} - 1\right)$

$i_2 = \frac{I_0}{3}\left[2e^{-\frac{3Rt}{2L}} + 1\right]$

The energy stored in inductance 'L' in steady state = $\frac{1}{2}L\left(\frac{I_0}{3}\right)^2 = \frac{LI_0^2}{18}$

The energy stored in inductance '2L' in steady state = $\frac{1}{2}2L\left(\frac{I_0}{3}\right)^2 = \frac{LI_0^2}{9}$

The net heat dissipated in the resistor = $\frac{1}{2}\times 3LI_0^2 - \left[\frac{LI_0^2}{18} + \frac{LI_0^2}{9}\right] = \frac{4}{3}LI_0^2$

7. D

8. A

Sol. (for Q.7-8)

Current over the surface of cylinder is

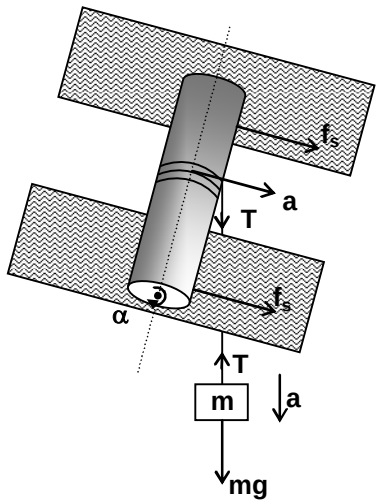
$I = \frac{Q\omega}{2\pi}$

$B = \frac{\mu_0 I}{\ell} = \frac{\mu_0 Q\omega}{2\pi\ell}$

Electric field induced on the surface of the cylinder is

$E = \frac{R}{2}\frac{dB}{dt} = \frac{R}{2}\frac{\mu_0 Q}{2\pi\ell}\frac{d\omega}{dt}$

$E = \frac{\mu_0 Q\alpha R}{4\pi\ell} = \frac{\mu_0 Qa}{4\pi\ell}$ …(i)



$mg - T = ma$ …(ii)

$TR - 2f_sR - QER = mR^2\alpha$

$T - 2f_s - QE = ma$ …(iii)

$2f_s = ma$ …(iv)

Adding (iii) and (iv), we get

$T - QE = 2ma$ …(v)

From (ii) and (v), we get

$mg - QE = 3ma$

$$mg - \frac{\mu_0 Q^2 a}{4\pi\ell} = 3ma$$

$mg - ma = 3ma$

$\therefore a = g/4$

Now from equation (ii)

$mg - T = ma$

$$T = mg - \frac{mg}{4} = \frac{3mg}{4}$$

9. C

10. D

Sol. (for Q.9-10)

$$\Delta Q_{AB} = nC_P\Delta T = 2 \times \frac{5R}{2}(900 - 300) = 3000R$$

In the process 'BC',

VT = constant

PV^2 = constant

$$\therefore \quad C = C_V + \frac{R}{1-x} = \frac{3R}{2} + \frac{R}{1-2} = \frac{R}{2}$$

$$\therefore \quad \Delta Q_{BC} = nC\Delta T = \frac{2R}{2}(300 - 900) = -600R$$

$$\Delta W_{BC} = \frac{nR\Delta T}{(1-x)} - \frac{2R(300-900)}{(1-2)} = 1200R$$

$$\Delta Q_{CA} = nRT_0 \ell n\left(\frac{V_0}{9V_0}\right) = 2R \times 300 \ell n\left(\frac{1}{9}\right) = -1200R\ell n(3) = -1200R \times 1.1 = -1320\ R$$

$$\therefore \quad \Delta W_{cycle} = \Delta Q_{cycle} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA} = 3000R - 600R - 1320R = 1080R$$

$$\therefore \quad \text{Efficiency of cycle, } \eta = \frac{\Delta W_{cycle}}{\Delta Q_{supplied}} \times 100 = \frac{1080R}{3000R} \times 100 = 36\%$$

SECTION – D

11. 00003.20

Sol. $v \cos\theta = v_0 \sin\theta$

$v = v_0 \tan\theta$ …(i)

$$a = v_0 \sec^2\theta \frac{d\theta}{dt}$$

$$a = v_0 \sec^2\theta\left(\frac{v_0\cos\theta + v\sin\theta}{\ell}\right) = \frac{v_0}{\ell}\sec\theta(v_0 + v\tan\theta)$$

$$a = \frac{v_0}{\ell}\sec\theta v_0(1 + \tan^2\theta)$$

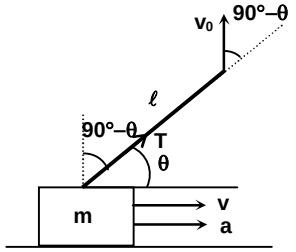
$$a = = \frac{v_0^2}{\ell}\sec^3\theta \quad \text{…(ii)}$$

$T\sin\theta = mg$ …(iii)

$T\cos\theta = ma$ …(iv)

From (iii) and (iv)

$g = a\tan\theta$



$$g = \frac{v_0^2}{\ell}\sec^3\theta\tan\theta$$

$$v_0 = \sqrt{\frac{g\ell}{\sec^3\theta\tan\theta}} = \left(\frac{4}{5}\right)^2\sqrt{\frac{10\times1.5\times5}{3}} = 3.20 \text{ m/s}$$

$v_0 = 3.20$ m/s

12. 00011.25

Sol. $N = mg\cos\theta$

$$f_k = \mu N = 2\tan\theta mg\cos\theta$$

$$f_k = 2mg\sin\theta$$

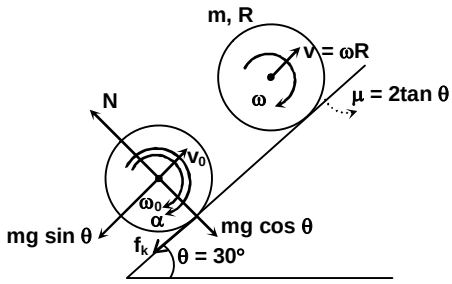
$$a = \frac{f_k + mg\sin\theta}{m} = 3g\sin\theta \qquad \ldots(i)$$

$\tau_{CM} = I_{CM}\alpha$

$$f_kR = \frac{2}{5}mR^2\alpha$$

$$\Rightarrow \quad 2mg\sin\theta = \frac{2}{5}m\alpha R$$

$$\alpha = \frac{5g\sin\theta}{R} \qquad \ldots(ii)$$



When pure rolling starts

$v = \omega R$

$v_0 - at = (\omega_0 + \alpha t)R$

$v_0 - 3g\sin\theta t = \omega_0 R + 5g\sin\theta t$

$v_0 - \omega_0 R = 8g\sin\theta t$

$$\therefore \ t = \frac{v_0}{12g\sin\theta}$$

$$\therefore \ v = v_0 - at = v_0 - 3g\sin\theta\times\frac{v_0}{12g\sin\theta} = v_0 - \frac{v_0}{4} = \frac{3v_0}{4}$$

$$\therefore \ v = \frac{3v_0}{4} = \frac{3\times15}{4} = 11.25 \text{ m/s}$$

13. 00010.50

Sol. $$\tan\theta = \frac{a + g\sin\alpha}{g\cos\alpha} = \frac{a + g\sin\alpha}{g\cos\alpha}$$

$$\frac{5-y}{3} = \frac{4+10\times\frac{3}{5}}{10\times\frac{4}{5}} \Rightarrow \frac{5-y}{3} = \frac{5}{4}$$

$\Rightarrow y = 1.25$ m

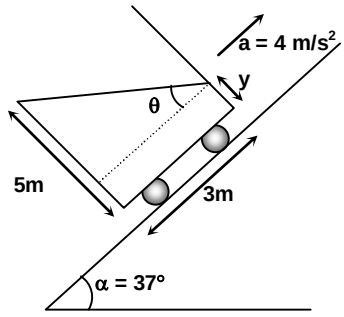
Volume of water remained in the container

$$V = \frac{1}{2}(5+1.25)\times3\times4 = 6.25\times6 = 37.50 \text{ m}^3$$

$\therefore$ volume of water spilt over the tank

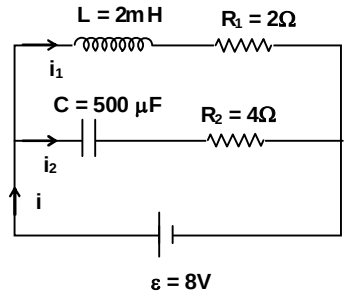
$\Delta V = V_0 - V$

$= (3\times4\times4) - 37.50 = 10.50 \text{ m}^3$



14. 00003.72

Sol. The current through the resistor 'R_1' is



$$i_1 = \frac{\varepsilon}{R_1}\left[1 - e^{-\frac{R_1 t}{L}}\right]$$

$$i_1 = 4\left[1 - e^{-\frac{t}{10^{-3}}}\right] \quad \ldots(i)$$

The current through the resistor 'R_2' is

$$i_2 = \frac{\varepsilon}{R_2} e^{-\frac{t}{R_2 C}}$$

$$i_2 = 2e^{-\frac{t}{2\times10^{-3}}} \quad \ldots(ii)$$

The current through the battery

$I = i_1 + i_2$

$$I = 4\left[1 - e^{\frac{-t}{10^{-3}}}\right] + \left(2e^{\frac{-t}{2\times10^{-3}}}\right)$$

At $t = 1 \times 10^{-3}$ sec

$I = 4(1 - e^{-1}) + 2e^{-0.5}$

$I = 4(1–0.37) + (2 \times 0.60) = 3.72$ amp

15. 00005.25

Sol. Emf induced in the straight segment AC is $\varepsilon_{AC} = 0$

Emf induced in the circular arc CD is

$$\varepsilon_{CD} = \int_a^{5a/2} B\omega r dr = \frac{1}{2}B\omega\left[r^2\right]_a^{\frac{5a}{2}} = \frac{21}{8}B\omega a^2$$

$$= \frac{21}{8}\times 5\times 10\times (0.2)^2 = 5.25 \text{ volt}$$

$$\therefore\ \varepsilon_{AD} = \varepsilon_{AC} + \varepsilon_{CD} = 0 + 5.25 = 5.25 \text{ volt}$$

16. 00003.74

Sol. Least count of screw gauge, $\text{L.C.} = \frac{1\,\text{mm}}{100} = 0.01\,\text{mm}$

Diameter of the wire, d = 2 mm + 46 × L.C. + 4 × L.C.

= 2mm + 46 × 0.01 mm + 4 × 0.01 mm = 2.50 mm = 0.250 cm

The length of wire, ℓ = 4.76 cm

$\therefore$ The curved surface area of the wire, $S = \pi d\ell$

$$= \frac{22}{7}\times 0.250 \times 4.76 = 3.74\,\text{cm}^2$$

17. 00025.60

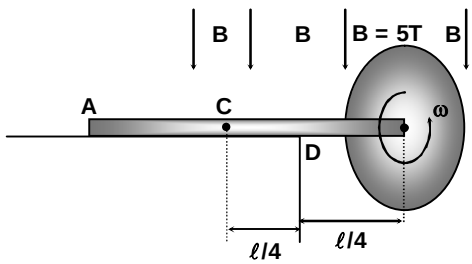
Sol. The magnetic moment of the disc is

$$M = \frac{q}{2m}\times\frac{mR^2\omega}{2} = \frac{q\omega R^2}{4} \quad \ldots(i)$$

Magnetic torque on the disc

$$\tau = MB = \frac{q\omega R^2 B}{4} \quad \ldots(ii)$$

For toppling of the rod about the edge at 'D'



$$\frac{q\omega R^2 B}{4} > (m_0 - m)\frac{g\ell}{4}$$

$$\omega > \frac{(m_0 - m)g\ell}{qBR^2}$$

$\omega > 25.60$

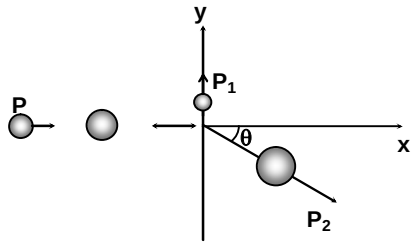
$\omega_{min} = 25.60$ rad/s

18. 00003.40

Sol. Using conservation of momentum

$P = P_2 \cos\theta$ …(i)

$P_1 = P_2 \sin\theta$ …(ii)



Squaring and adding, we get

$P^2 + P_1^2 = P_2^2$

$2mk + 2m_1k_1 = 2m_2K_2$

$4k + k_1 = 10k_2$

$10k_2 - k_1 = 4k$

$\therefore$ $10k_2 - k_1 = 44.$ …(iii)

Also, $k_1 + k_2 = k - |Q|$

$k_1 + k_2 = 11. - 2.86$

$k_1 + k_2 = 8.14$ …(iv)

Adding (iii) and (iv), we get

$11k_2 = 52.14$

$\therefore$ $k_2 = 4.74$ MeV

$\therefore$ $k_1 = 8.14 - 4.74$

$k_1 = 3.40$ MeV

Chemistry PART – II

SECTION – A

19. B, C, D

Sol. (A) For real gas $\frac{dU}{dV} = \frac{an^2}{V^2}$ i.e. internal energy is volume dependent also.

(B) $Z > 1 \Rightarrow V_{real} > V_{ideal}$ so forces of repulsion are dominant.

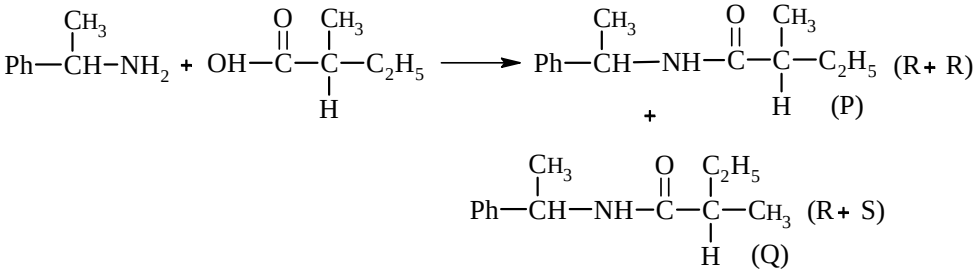
(D) For ideal gas $\Delta U = 0$ but for real gas $\Delta U \neq 0$.

20. A, C

Sol. 1^0 amines give carbylamines reaction

21. A, B

Sol.



22. A, D

Sol. (D) The reducing character of hydrides of 15 group elements increases from NH_3 to BiH_3.

23. A, C, D

Sol. As Ni is added, CO will be consumed to attain the third equilibrium. As concentration of CO decreases, $COCl_2$ will decompose to a higher extent and hence Cl_2 concentration increases which in turn decreases the concentration of PCl_3.

24. B, C

Sol. In $F_2C = C = CF_2$

Terminal C-atoms are sp^2 hybridised, one sp^2 hybridised orbital comes in the horizontal plane and other sp^2 hybridised orbital appears on the vertical plane.

In N_2H_4, the nitrogen atoms are sp^3 hybridised.

25. B

Sol. Meq of $H_3PO_4 = 300 \times 0.1 = 30$

Meq of $NaOH = 50 \times 0.1 = 5$

$H_3PO_4 + NaOH = NaH_2PO_4 + H_2O$

30	5	0	0
25	0	5	5

Solution is buffer

$pH = P^{ka_1} + \log \frac{5}{25} \Rightarrow 3 + \log 1/5$

$\Rightarrow$ 2.3010

26. A

Sol. At point 'C' a buffer of H_2 PO_4^- & HPO_4^{2-} exists.

27. A

Sol. $pI = \frac{pKa_1 + pKa_2}{2}$

Isoelectric point (PI) $= \frac{pK_{a_1} + pK_{a_2}}{2}$

$= \frac{1.9 + 3.7}{2} = \frac{5.6}{2} = 2.8$

28. A

Sol. $H_2N–CH_2–COOH$ $\xrightarrow{pH = 2}$ $\overset{\oplus}{N}H_3–CH_2–COOH$ Cationic (will move towards cathode)

$H_2N–CH_2–COOH$ $\xrightarrow{pH = 9}$ $H_2NCH_2COO^-$ Anionic $\longrightarrow$ will move towards anode

SECTION – D

29. 00001.46

Sol. Suppose relative reactivity for 3° H is x
and suppose relative reactivity for 1° H is Y
So with the given information

$\frac{9y}{9y + x} = 0.86 \frac{x}{9y + x} = 0.14$

So $\frac{9y}{0.86} = \frac{x}{0.14}$

Hence, $\frac{x}{y} = \frac{9 \times 0.14}{0.86} = 1.46$

30. 00023.00

Sol. Each 111 g of $CaCl_2$ ≡ 100 g of $CaCO_3$
So 55.5 g of $CaCl_2$ ≡ 50 g of $CaCO_3$
Each 95 g of $MgCl_2$ ≡ 100 g of $CaCO_3$
So 47.5 of $MgCl_2$ ≡ 50 g of $CaCO_3$
Each 120 g of $MgSO_4$ ≡ 100 g of $CaCO_3$
So 84 g of $MgSO_4$ ≡ 70 g of $CaCO_3$
Each 136 g of $CaSO_4$ ≡ 100 g of $CaCO_3$
So 81.6 g of $CaSO_4$ ≡ 60 g of $CaCO_3$
Hence total $CaCO_3$ = 230 g

Degree of hardness $= \frac{230 \text{ g}}{10,000 \ell\text{t}} = 23 \text{ mg / litre} = 23 \text{ ppm.}$.

31. 00002.81

Sol. $K = \frac{0.693}{12}$

$t = \frac{2.303}{0.693} \times t_{1/2} \log \frac{1}{0.85}$

t = 2.81 mins.

32. 00002.64

Sol. $z = 4 - \frac{4 \times 0.25}{100} = 3.99$

$\rho = \frac{z \times M}{a^3 \times N_A \times 10^{-30}} g/cc$

$= \frac{3.99 \times 50}{(500)^3 \times 6.023 \times 10^{23} \times 10^{-30}} g/cc$

$= 2.64\ g/cc$

33. 00006.44

Sol. $w = \frac{E \times I \times t}{96500}$

$w = d \times V$

$= 19.3\ g\ cm^{-3} \times \pi \times 2.5 \times 2.5 \times 0.02\ cm^{-3}$

$w = 7.575\ g$

$t = \frac{7.575 \times 96500 \times 3}{197 \times 0.02}$

$= 556589.5\ sec$

≈ 6.44 days

34. 00083.50

Sol. $2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(\ell)\ \ \Delta H = -3129KJ$

$\Delta H = \sum \Delta H_{f(Product)} - \sum \Delta H_{f(reactants)}$

$-3129 = \left[4 \times (-395) + 6 \times -(286)\right] - \left[2 \times \Delta H_f C_2H_6 + 7 \times 0\right]$

$\Delta H_{fC_2H_6} = -83.5KJ$

35. 00014.64

Sol.

$$\text{No. of equivalent of } Na_2S_2O_3 = \frac{100 \times 0.15}{1000} = 0.015$$

Let mass of $K_2Cr_2O_7 = a$

Mass of $KMnO_4 = 0.5 - a$

$$\therefore \text{No of equivalent of } K_2Cr_2O_7 = \frac{a}{49}$$

$$\text{No. of equivalent of } KMnO_4 = = \frac{0.5 - a}{31.6}$$

$$\therefore \frac{a}{49} + \frac{0.5 - a}{31.6} = 0.015$$

$a = 0.0732$

$$\% \text{ of } K_2Cr_2O_7 = \frac{0.0732}{0.5} \times 100$$

$= 14.64\%$

36. 00001.48

Sol.

$$\frac{r_{mix}}{r_{SO_2}} = \sqrt{\frac{M_{SO_2}}{M_{mix}}}, \quad 1.6 = \sqrt{\frac{M_{SO_2}}{M_{mix}}}$$

$M_{mix} = 25$

Let mole fraction of F_2 be 'x'

$$25 = \frac{38x + (1-x) \times 19}{1}$$

$X = 0.3158$

$$K_P = \frac{(0.6842P)^2}{0.3158P} = 1.48 \text{ atm } (P = 1)$$

Mathematics PART – III

SECTION – A

37. A, C

Sol. Equation of AB $\Rightarrow y = \frac{4}{3}(x+1)$

AC $\Rightarrow y = -\frac{4}{3}(x-1)$

BC $\Rightarrow y = 0$

Distance of P from AB = $d_1 = \frac{1}{5}|4x-3y+4|$

Distance of P from AC = $d_2 = \frac{1}{5}|4x+3y-4|$

Distance of P from BC = $d_3 = |y|$

According to question $d_1d_2 = d_3^2$

$\Rightarrow |16x^2 - (3y-4)^2| = 25y^2$

$\Rightarrow$ Circle S: $2x^2 + 2y^2 + 3y - 2 = 0$ and hyperbola T: $8x^2 - 17y^2 + 12y - 8 = 0$

38. A, C, D

Sol. Let $f(x) = (1+x)(1+x^2) \ldots (1+x^{2070})$

Then A must be the sum of coefficient of x^{9k} where k is non-negative integer

$A = \frac{2^{2070} + 8.2^{230}}{9}$

39. B, C

Sol. Let $a = a_1 + ia_2$; $c = c_1 + ic_2$, then $\frac{a_1}{a_2} \times \frac{c_1}{c_2} = -1$

$\Rightarrow a_1c_1 + a_2c_2 = 0$

$\Rightarrow \left(\frac{a+\bar{a}}{2}\right)\left(\frac{c+\bar{c}}{2}\right) + \left(\frac{a-\bar{a}}{2i}\right)\left(\frac{c-\bar{c}}{2i}\right) = 0$

$\Rightarrow a\bar{c} + \bar{a}c = 0$

$\therefore$ $a\bar{c}$ is purely imaginary

Also $\frac{a}{c} = -\frac{\bar{a}}{\bar{c}}$

$\Rightarrow \frac{a}{c}$ is purely imaginary

$\Rightarrow \arg\left(\frac{a}{c}\right) = \pm\frac{\pi}{2}$

40. A, B

Sol. ${}^nC_7 = \frac{n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)}{7!}$

Obviously one of the seven numerator factors must be divisible by 49.

Of any 49 consecutive integer exactly seven satisfy the condition $P\left(\frac{{}^nC_7}{7}\right) = \frac{7}{49} = \frac{1}{7}$

$${}^{n}C_7 = \frac{n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)}{7!}$$

So for $\frac{{}^{n}C_7}{7}$ to be divisible by 12

$S = n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)$ must be divisible by $= 64.27 \Rightarrow 2^6 \cdot 3^3$

Any one of the seven factors must be divisible by 9. So, here only 7 favourable choices

Also, if n is even then S is divisible by 64 and if n is odd then one after the (n – 1), (n – 3), (n – 5) must be divisible by 16

So we have 13 favourable choices out of 16 consecutive numbers

Required probability that ${}^{n}C_7$ divisible by 12 $= \frac{7 \times 13}{144}$

41. B, C

Sol. $\Delta = 16\cos^2\theta - 4\cot\theta = 0$

or $4\cot\theta(2\sin 2\theta - 1) = 0$

$2\sin 2\theta - 1 = 0$

$$\theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}$$

42. A, B, C, D

Sol. $$I = \frac{1}{2}\int_0^{\pi} 2\cos mx \cdot \sin nx\,dx = \frac{1}{2}\int_0^{\pi}\left[\sin(m+n)x + \sin(n-m)x\right]dx$$

$$= -\frac{1}{2}\left[\left\{\frac{(-1)^{m+n}}{m+n} + \frac{(-1)^{n-m}}{n-m}\right\} - \left\{\frac{1}{m+n} + \frac{1}{n-m}\right\}\right] \text{ for } k \in I \left[\because \cos k\pi = (-1)^k\right]$$

$$\Rightarrow I = \begin{cases} \frac{2n}{n^2-m^2} & \text{If } n-m \text{ is odd} \\ 0 & \text{If } n-m \text{ is even} \end{cases}$$

43. A

44. B

Sol. (43.-44.) $\left(x\hat{i} + y\hat{j} + z\hat{k}\right)\cdot\left(\hat{i} + \hat{j}\right) = 2$

$x + y = 2$ ..... plane π_1

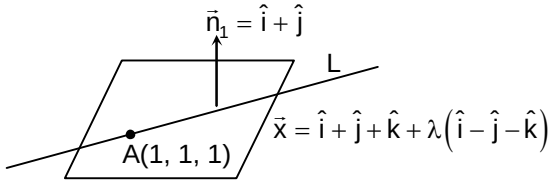
$$\overrightarrow{AB} \times \vec{V} = \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{vmatrix}$$

$\vec{n}_2 = \hat{i} + 2\hat{j} - \hat{k}$

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

$$\cos\theta = \left|\frac{\left(\hat{i} + \hat{j}\right)\cdot\left(\hat{i} + 2\hat{j} - \hat{k}\right)}{\sqrt{2}\,\sqrt{6}}\right| = \frac{3}{\sqrt{2}\cdot\sqrt{6}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \cos^{-1}\frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$



45. B

46. A

Sol. (45.-46.) $|z_1| = 2$

$$3\left(\frac{z_1}{z_2}\right)^2 - 2\left(\frac{z_1}{z}\right) + 2 = 0$$

$$\frac{z_1}{z_2} = \frac{2 \pm \sqrt{-20}}{6} = \frac{1 \pm \sqrt{5}\,i}{2}$$

$$\frac{|z_1|}{|z_2|} = \frac{\sqrt{6}}{3} = \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow |z_2| = \sqrt{6}$$

$a = 2,\ b = \sqrt{6}$

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{1}{\sqrt{3}}$$

Let $\frac{z_1}{z_2} = \frac{1}{3} + \frac{\sqrt{5}}{3}i = \frac{\sqrt{6}}{3}\left(\frac{1+\sqrt{5}}{\sqrt{6}}i\right)$

$$\sin\theta = \frac{\sqrt{5}}{\sqrt{6}}$$

$$\Delta OPQ = \frac{1}{2}|z_1|\,|z_2|\sin\theta = \frac{1}{2} \times 2 \times \sqrt{6} \times \frac{\sqrt{5}}{\sqrt{6}} = \sqrt{5}$$

SECTION – D

47. 02021.00

Sol. $$\lim_{n\to\infty} n^2 \int_{-\frac{1}{n}}^{\frac{1}{n}} (2019\sin x + 2021\cos x)|x|\,dx = \lim_{n\to\infty} n^2 \cdot 4042 \int_0^{\frac{1}{n}} x\cos x\,dx$$

$$= \frac{4042 \times \frac{1}{n}\cos\left(\frac{1}{n}\right)\left(-\frac{1}{n^2}\right)}{\frac{2}{n^3}} = 2021$$

48. 02000.00

Sol. Let $\vec{\alpha} = 3\hat{i} + 4\hat{j} + 10\hat{k}$ and $\vec{\beta} = a\hat{i} + b\hat{j} + c\hat{k}$

$$S = \vec{\alpha}\cdot\vec{\beta} = |\vec{\alpha}||\vec{\beta}|\cos\theta \le |\vec{\alpha}||\vec{\beta}| \le \sqrt{2000}$$

49. 00021.00

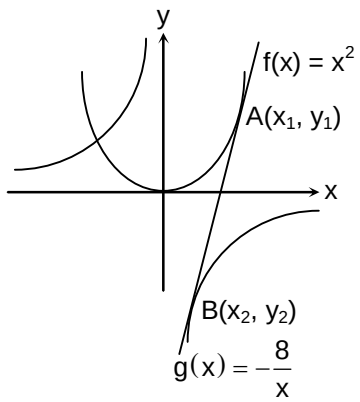
Sol. $y = x^2$ and $y = -\frac{8}{x}$

$$\Rightarrow y_1 = x_1^2 \text{ and } y_2 = -\frac{8}{x_2}$$

Equality $\frac{dy}{dx}$ at A and B, we get $2x_1 = \frac{8}{x_2^2}$

$$\Rightarrow x_1 x_2^2 = 4$$

$$m_{AB} \Rightarrow 2x_1 = \frac{x_1^2 + \frac{8}{x_2}}{x_1 - x_2}$$



$\Rightarrow x_1 = 4$ and $x_2 = 1$, $y_1 = 16$ $\therefore x_1 + y_1 + x_2 = 21$

50. 00503.50

Sol. From the relation between the zeros and the coefficient we obtain

$\sum x_1x_2.....x_{1007} = 1008$ and x_1x_2 ..... $x_{1008} = 1$

Hence, $\sum_{i=1}^{1008} \frac{1}{x_i} = 1008$

So we have the equality case in the A.M. – G.M. inequality

So $x_1 = x_2$ ..... $= x_{1008} = 1$

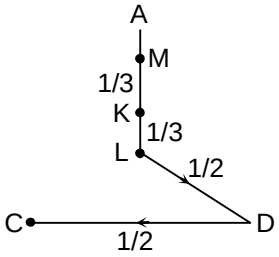
$\Rightarrow a_2 = \frac{1008^2 - 1008}{2}$; $a_1 = -1008$

So $\frac{a_2}{|a_1|} = 503.50$

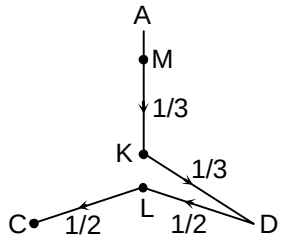
51. 00008.00

Sol. Possible path mouse can reach C

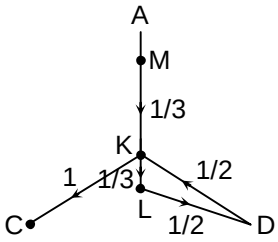
(i)



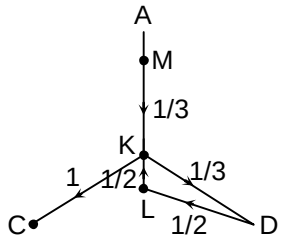
(ii)



(iii)



(iv)



Hence the required probability

$= \frac{1}{3}\times\frac{1}{3}\times\frac{1}{2}\times\frac{1}{2}+\frac{1}{3}\times\frac{1}{3}\times\frac{1}{2}\times\frac{1}{2}+\frac{1}{3}\times\frac{1}{3}\times\frac{1}{2}\times\frac{1}{2}\times 1+\frac{1}{3}\times\frac{1}{3}\times\frac{1}{2}\times\frac{1}{2}\times 1$

$= \frac{1}{9}$

52. 00071.00

Sol. $\sin\frac{\pi}{7}\sin\frac{2\pi}{7}\sin\frac{3\pi}{7}\sin\frac{4\pi}{7}\sin\frac{5\pi}{7}\sin\frac{6\pi}{7}$

$\Rightarrow \left(\sin\frac{\pi}{7}\sin\frac{2\pi}{7}\sin\frac{4\pi}{7}\right)^2$

Let $7\theta = \pi$

$\Rightarrow 3\theta + 4\theta = \pi$

$\Rightarrow 4\theta = \pi - 3\theta$

$\Rightarrow \sin 4\theta = \sin(\pi - 3\theta)$

$\Rightarrow \sin 4\theta = \sin 3\theta$

$\Rightarrow 4 \sin\theta \cos\theta(\cos 2\theta) = 3 \sin\theta - 4 \sin^3\theta$

$\Rightarrow 16 \cos^2\theta(\cos 2\theta)^2 = (3 - 4 \sin^2\theta)^2$

$\Rightarrow 16(1 - \sin^2\theta)(1 - 2 \sin^2\theta)^2 = (3 - 4 \sin^2\theta)^2$

$\Rightarrow -64\sin^6\theta + 80\sin^4\theta - 80\sin^2\theta + 16 = 16\sin^4\theta + 9 - 24\sin^2\theta$

$\Rightarrow 64\sin^6\theta - 66\sin^4\theta + 56\sin^2\theta - 7 = 0$

$\Rightarrow \left(\sin\frac{\pi}{7}\sin\frac{2\pi}{7}\sin\frac{3\pi}{7}\right)^2 = \frac{7}{64}$

53. 00002.50

Sol. $y^2 = x^2 - x + 1 \Rightarrow 2y\frac{dy}{dx} = 2x - 1$

$AI_3 + BI_2 + CI_1 = x^2y$

$\Rightarrow \int \frac{Ax^3 + Bx^2 + Cx}{y}dx = x^2y$

$\Rightarrow \frac{Ax^3 + Bx^2 + Cx}{y} = x^2\frac{dy}{dx} + 2xy$

$\Rightarrow Ax^3 + Bx^2 + Cx = 3x^3 - \frac{5x^2}{2} + 2x$

$\Rightarrow A = 3, B = -\frac{5}{2}, C = 2$

So, $A + C = 5$

54. 00007.00

Sol. Substitute $t = \sin x$ and $g(t) = t + \frac{2}{3+t} + b$

Since g(t) is an increasing function in the interval [–1, 1]

It follows that $f(b) = \max(|g(-1)|, |g(1)|) = \max\left(|b|, \left|b + \frac{3}{2}\right|\right)$

From the graph we conclude that $\min f(b) = f\left(-\frac{3}{4}\right) = \frac{3}{4}$

So $p + q = 7$